Complete Marketing Mix Model Parameter Guide

Channel-Specific Response Parameters

Ad Network 1 (ADNW1)

- **ADNW1_alphas**: Saturation parameter - controls diminishing returns curve
- **ADNW1_gammas**: Lag parameter - controls delay before effects begin
- **ADNW1_thetas**: Adstock parameter - controls how long effects persist

Ad Network 2 (ADNW2)

- **ADNW2_alphas**: Saturation parameter
- **ADNW2_gammas**: Lag parameter
- **ADNW2_thetas**: Adstock parameter

Ad Network 3 (ADNW3)

- **ADNW3_alphas**: Saturation parameter
- **ADNW3_gammas**: Lag parameter
- **ADNW3_thetas**: Adstock parameter

Affiliate Marketing (AFF)

- **AFF_alphas**: Saturation parameter
- **AFF_gammas**: Lag parameter
- **AFF_thetas**: Adstock parameter

Brand Marketing (BRAND)

- **BRAND_alphas**: Saturation parameter
- **BRAND_gammas**: Lag parameter
- **BRAND_thetas**: Adstock parameter

Video Advertising (DOUGA)

- **DOUGA_alphas**: Saturation parameter
- **DOUGA_gammas**: Lag parameter
- **DOUGA_thetas**: Adstock parameter

Game Promotion - March 24, 2023 (GAMEPR230324)

- **GAMEPR230324_alphas**: Saturation parameter
- **GAMEPR230324_gammas**: Lag parameter
- **GAMEPR230324_thetas**: Adstock parameter

Game Promotion - July 27, 2023 (GAMEPR230727)

- **GAMEPR230727_alphas**: Saturation parameter
- **GAMEPR230727_gammas**: Lag parameter
- **GAMEPR230727_thetas**: Adstock parameter

Influencer Campaign - April 2, 2023 (INFLU230402)

- **INFLU230402_alphas**: Saturation parameter
- **INFLU230402_gammas**: Lag parameter
- **INFLU230402_thetas**: Adstock parameter

Influencer Campaign - April 4, 2023 (INFLU230404)

- **INFLU230404_alphas**: Saturation parameter
- **INFLU230404_gammas**: Lag parameter
- **INFLU230404_thetas**: Adstock parameter

Influencer Campaign - April 8, 2023 (INFLU230408)

- **INFLU230408_alphas**: Saturation parameter
- **INFLU230408_gammas**: Lag parameter
- **INFLU230408_thetas**: Adstock parameter

Influencer Campaign - August 21, 2023 (INFLU230821)

- **INFLU230821_alphas**: Saturation parameter
- **INFLU230821_gammas**: Lag parameter
- **INFLU230821_thetas**: Adstock parameter

Search Engine Marketing (SEM)

- **SEM_alphas**: Saturation parameter
- **SEM_gammas**: Lag parameter
- **SEM_thetas**: Adstock parameter

Social Network Sites (SNS)

- **SNS_alphas**: Saturation parameter
- **SNS_gammas**: Lag parameter
- **SNS_thetas**: Adstock parameter

Model Performance Metrics

R-squared Values

- **rsq_train**: R-squared for training data - higher is better
- **rsq_val**: R-squared for validation data - measures generalization
- **rsq_test**: R-squared for test data - measures out-of-sample performance

Error Metrics

- **nrmse_train**: Normalized RMSE for training data
- **nrmse_val**: Normalized RMSE for validation data
- **nrmse_test**: Normalized RMSE for test data
- **nrmse**: Overall normalized RMSE
- **decomp.rssd**: Decomposition residual sum of squared deviations
- **mape**: Mean Absolute Percentage Error
- **mape.qt10**: 10th quantile of MAPE values
- **error_score**: Consolidated error measurement

Regularization & Hyperparameters

- **lambda**: Regularization strength parameter
- **lambda_hp**: Hyperparameter for lambda
- **lambda_max**: Maximum lambda value considered
- **lambda_min_ratio**: Ratio of minimum to maximum lambda
- **train_size**: Proportion of data used for training
- **coef0**: Initial coefficient value

Solution Identification & Runtime

- **pos**: Position in model ranking
- **solID**: Solution identifier
- **trial**: Trial number
- **cluster**: Cluster grouping for similar solutions
- **top_sol**: Indicator for top solution
- **robynPareto**: Pareto-efficient solution indicator
- **Elapsed**: Time taken for this solution
- **ElapsedAccum**: Accumulated time across solutions

Iteration Details

- **iterNG**: Number of gradient-based iterations
- **iterPar**: Number of parameter iterations
- **iterations**: Total iterations

Understanding Parameter Importance

1. **Response Parameters (alphas, gammas, thetas)**:
    - Together define how each marketing channel impacts sales
    - Higher thetas mean longer-lasting effects after exposure
    - Higher alphas indicate quicker diminishing returns
    - Higher gammas show greater delay before seeing effects
2. **Performance Metrics**:
    - R-squared values closer to 1 indicate better model fit
    - Lower NRMSE and MAPE values indicate more accurate predictions
    - Check for big differences between train/val/test to identify overfitting
3. **Regularization Parameters**:
    - Control model complexity to prevent overfitting
    - Optimize balance between fitting training data and generalizing to new data

These parameters collectively determine how well your model captures marketing effects, how quickly different channels drive results, and how reliably you can use the model for budget optimization.